

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$y = 18$$

$$y = -3(x - 18)^2 + 15$$

If the given equations are graphed in the xy -plane, at how many points do the graphs of the equations intersect?

Answer

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Correct Answer: D

Rationale

Choice D is correct. A point (x, y) is a solution to a system of equations if it lies on the graphs of both equations in the xy -plane. In other words, a solution to a system of equations is a point (x, y) at which the graphs intersect. It's given that the first equation is $y = 18$. Substituting 18 for y in the second equation yields $18 = -3(x - 18)^2 + 15$. Subtracting 15 from each side of this equation yields $3 = -3(x - 18)^2$. Dividing each side of this equation by -3 yields $-1 = (x - 18)^2$. Since the square of a real number is at least 0 , this equation can't have any real solutions. Therefore, the graphs of the equations intersect at zero points.

Alternate approach: The graph of the second equation is a parabola that opens downward and has a vertex at $(18, 15)$. Therefore, the maximum value of this parabola occurs when $y = 15$. The graph of the first equation is a horizontal line at 18 on the y -axis, or $y = 18$. Since 18 is greater than 15 , or the horizontal line is above the vertex of the parabola, the graphs of these equations intersect at zero points.

Choice A is incorrect. The graph of $y = 15$, not $y = 18$, and the graph of the second equation intersect at exactly one point.

Choice B is incorrect. The graph of any horizontal line such that the value of y is less than 15 , not greater than 15 , and the graph of the second equation intersect at exactly two points.

Choice C is incorrect and may result from conceptual or calculation errors.